LAB	PRACTICALS
1	**Introduction to Database, Database Tools/Editors, SQL and Components of SQL**

1. WHY DATABASE? ADVANTAGES OF DATABASE

Database

- A Database is an organized collection of related data stored electronically. It allows users to store, retrieve, update, and manage data efficiently.
- Database software makes data management simpler by enabling users to store data in a structured form and then access it. It typically has a graphical interface to help create and manage the data and, in some cases, users can construct their own databases by using database software.

Advantages of Database

- **Data Consistency** – Avoids duplicate and conflicting data
- **Data Security** – Restricts unauthorized access
- **Data Sharing** – Multiple users can access data simultaneously
- **Reduced Data Redundancy** – Eliminates duplicate data
- **Data Integrity** – Ensures accuracy and reliability of data
- **Backup and Recovery** – Protects data from loss
- **Centralized Control** – Easy data management
- **Scalability** – Handles growing data efficiently

2. DIFFERENT TYPES OF DATABASES & TOOLS/EDITORS AVAILABLE FOR IT.

Types of Databases:

- Hierarchical Database
- Network Database
- Relational Database (RDBMS)
- Object-Oriented Database
- Entity-Relational Database
- Client-Server Database
- Distributed Database
- NoSQL Database
- Cloud Database

Databases Tools / Editors:

- MySQL
- SQL Server
- Oracle
- SQLite
- PostgreSQL
- IBM Db2
- MongoDB
- Microsoft Access

3. WHAT IS SQL? COMPONENTS OF SQL (DDL, DML, DCL, DQL, TCL)

SQL

- SQL (Structured Query Language) is a standard language used to communicate with relational databases. It is used to create, retrieve, update, delete, and control database data.

Components of SQL

1. **DDL – Data Definition Language**

Command	Description
CREATE	Creates database/table
ALTER	Modifies table structure
DROP	Deletes table/database
TRUNCATE	Removes all table records
RENAME	Rename table/columns

2. **DML – Data Manipulation Language**

Command	Description
INSERT	Adds new records
UPDATE	Modifies existing records
DELETE	Removes records

3. **DQL – Data Query Language**

Command	Description
SELECT	Retrieve data

4. **DCL – Data Control Language**

Command	Description
GRANT	Gives user to access rights to database/table
REVOKE	Removes/withdraw user access rights given by using GRANT command.

5. **TCL – Transaction Control Language**

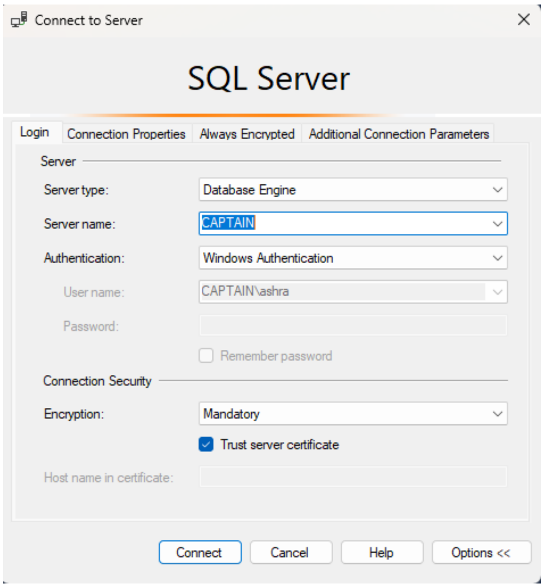
Command	Description
COMMIT	Saves transaction permanently
ROLLBACK	Rollback/Undo changes
SAVEPOINT	Sets restore point within a transaction

2 **Introduction to SQL Server Management Studio, Database, Table, Fields, Records, Data types in SQL Server**

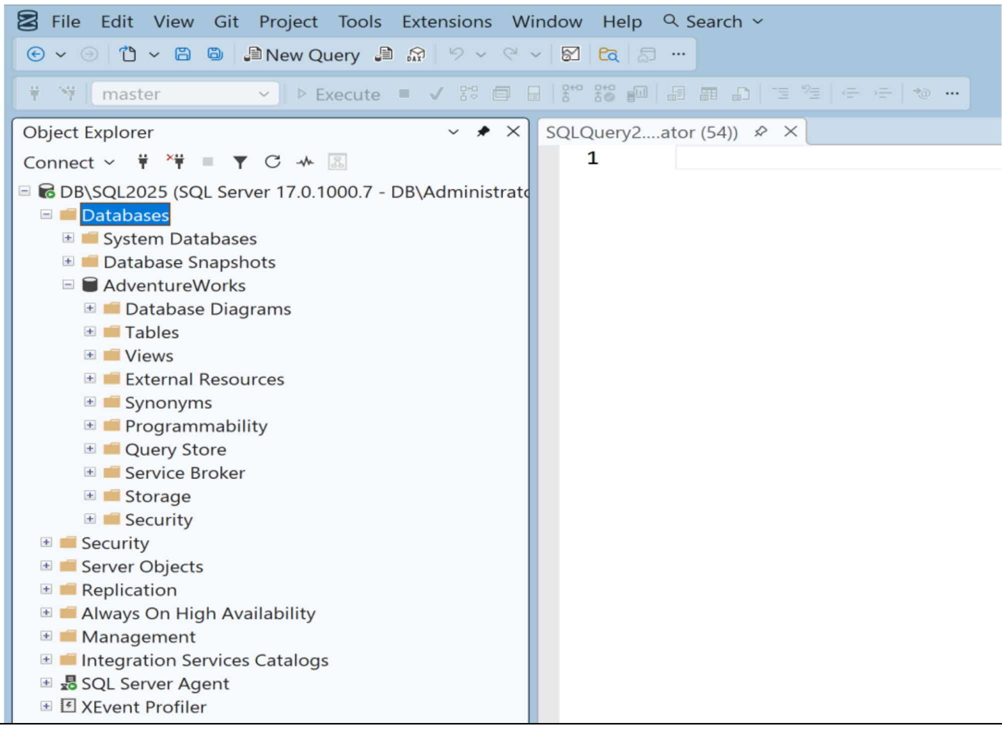
1. Introduction to Editor (SQL Server Management Studio)

- SQL Server Management Studio (SSMS) is an essential tool for effectively managing SQL Server databases. Developed by Microsoft, SSMS provides a comprehensive, integrated environment where database administrators, developers, and other users can efficiently work with their SQL Server instances.
- Install SQL Server Management Studio and SQL Server from below link:

- [Microsoft SQL Server Downloads]
- Once the installation is complete, SQL Server Management Studio opens as shown below.
- As soon as we open SSMS, the *Connect to Server* dialog box appears as shown below:



- Enter the server name and select the appropriate authentication method for the SQL Server you want to connect to, then click the *Connect* button to connect to SQL Server Management Studio. If MS SQL Server was installed using the default settings, the server name will be the same as the one specified during the installation of MS SQL Server.
- After connecting to the database, the SQL Server Management Studio interface appears as shown below:



2. Introduction to Database, Table, Field, Row, Record.

Database

- A database is an organized collection of related data stored electronically in a computer system. It allows users to store, retrieve, update, and manage data efficiently. Databases reduce data redundancy and improve data security and consistency.

Table

- A table is a database object used to store data in a structured format of rows and columns. Each table represents a specific entity, such as *Student*, *Employee*, or *Product*. Tables make data easy to view and manage.

Field

- A field is a single column in a table. It represents a specific attribute of an entity. For example, *StudentID*, *Name*, and *Age* are fields in a Student table. Each field stores a particular type of data.

Row

- A row is a single horizontal entry in a table. It contains data related to one instance of an entity. For example, one row in a Student table contains complete information about one student.

Record

- A record is a complete set of fields related to one entity, stored in a single row of a table. In simple terms, a record and a row represent the same concept.

3. Introduction to various data types in SQL: INT, CHAR, VARCHAR, DATETIME, BIT, DECIMAL.

- SQL data types define the type of data that can be stored in a table column. Choosing the correct data type helps in efficient storage and accurate data processing.

Data Type	Description	Example	Common Usage
INT	Stores whole numbers without decimal values	10, -25	Roll number, Employee ID
CHAR(n)	Stores fixed-length character data	CHAR(10)	Gender, fixed-size codes
VARCHAR(n)	Stores variable-length character data	VARCHAR(50)	Name, Email address
DATETIME	Stores date and time values	2025-01-01 10:30:00	Joining date, Order date
BIT	Stores Boolean values (0 or 1)	1 (True), 0 (False)	Status (Active/Inactive)
DECIMAL(p,s)	Stores exact numeric values with precision and scale	DECIMAL(10,2)	Salary, Price

3 **Implementation of the CREATE and INSERT commands in SQL**
Create a database with the name: YourName_EnrolmentNo
Create the following tables under your database and insert the data using Query Mode.

```sql
CREATE TABLE DEPOSIT
(
	ACTNO INT,
	CNAME VARCHAR(50),
	BNAME VARCHAR(50),
	AMOUNT DECIMAL(8,2),
	ADATE DATETIME
);

CREATE TABLE BRANCH
(
	BNAME VARCHAR(50),
	CITY VARCHAR(50)
);

CREATE TABLE CUSTOMERS
(
	CNAME VARCHAR(50),
	CITY VARCHAR(50)
);

CREATE TABLE BORROW
(
	LOANNO INT,
	CNAME VARCHAR(50),
	BNAME VARCHAR(50),
	AMOUNT DECIMAL(8,2)
);
```

DEPOSIT

```sql
INSERT INTO DEPOSIT VALUES(101,'ANIL','VRCE',1000.00,'1/MAR/95')
INSERT INTO DEPOSIT VALUES(102,'SUNIL','AJNI',5000.00,'4/JAN/96')
INSERT INTO DEPOSIT VALUES(103,'MEHUL','KAROLBAGH',3500.00,'17/NOV/95')
INSERT INTO DEPOSIT VALUES(104,'MADHURI','CHANDI',1200.00,'17/DEC/95')
INSERT INTO DEPOSIT VALUES(105,'PRMOD','M.G.ROAD',3000.00,'27/MAR/96')
INSERT INTO DEPOSIT VALUES(106,'SANDIP','ANDHERI',2000.00,'31/MAR/96')
INSERT INTO DEPOSIT VALUES(107,'SHIVANI','VIRAR',1000.00,'5/SEP/95')
INSERT INTO DEPOSIT VALUES(108,'KRANTI','NEHRU PLACE',5000.00,'2/JUL/95')
INSERT INTO DEPOSIT VALUES(109,'MINU','POWAI',7000.00,'10/AUG/95')
INSERT INTO DEPOSIT VALUES(110,'KARAN','BANDRA',4500.00,'23/FEB/94')
```

BRANCH

```sql
INSERT INTO BRANCH VALUES('VRCE','NAGPUR')
INSERT INTO BRANCH VALUES('AJNI','NAGPUR')
INSERT INTO BRANCH VALUES('KAROLBAGH','DELHI')
```

	INSERT INTO BRANCH VALUES('CHANDI','DELHI') INSERT INTO BRANCH VALUES('DHARAMPETH','NAGPUR') INSERT INTO BRANCH VALUES('M.G.ROAD','BANGLORE') INSERT INTO BRANCH VALUES('ANDHERI','BOMBAY') INSERT INTO BRANCH VALUES('VIRAR','BOMBAY') INSERT INTO BRANCH VALUES('NEHRU PLACE','DELHI') INSERT INTO BRANCH VALUES('POWAI','BOMBAY') **CUSTOMERS** INSERT INTO CUSTOMERS VALUES('ANIL','CULCUTTA') INSERT INTO CUSTOMERS VALUES('SUNIL','DELHI') INSERT INTO CUSTOMERS VALUES('MEHUL','BARODA') INSERT INTO CUSTOMERS VALUES('MANDAR','PATNA') INSERT INTO CUSTOMERS VALUES('MADHURI','NAGPUR') INSERT INTO CUSTOMERS VALUES('PRMOD','NAGPUR') INSERT INTO CUSTOMERS VALUES('SANDIP','SURAT') INSERT INTO CUSTOMERS VALUES('SHIVANI','BOMBAY') INSERT INTO CUSTOMERS VALUES('KRANTI','BOMBAY') INSERT INTO CUSTOMERS VALUES('NAREN','BOMBAY') **BORROW** INSERT INTO BORROW VALUES(201,'ANIL','VRCE',1000) INSERT INTO BORROW VALUES(206,'MEHUL','AJNI',5000) INSERT INTO BORROW VALUES(311,'SUNIL','DHARAMPETH',3000) INSERT INTO BORROW VALUES(321,'MADHURI','ANDHERI',2000) INSERT INTO BORROW VALUES(375,'PRMOD','VIRAR',8000) INSERT INTO BORROW VALUES(481,'KRANTI','NEHRU PLACE',3000)
4	**Implementation of the SELECT command in SQL** **From the tables given in Lab-3, perform the following queries using the SELECT command:** **Part - A** 1. RETRIEVE ALL DATA FROM TABLE DEPOSIT. SELECT * FROM DEPOSIT 2. RETRIEVE ALL DATA FROM TABLE BORROW. SELECT * FROM BORROW 3. RETRIEVE ALL DATA FROM TABLE CUSTOMERS. SELECT * FROM CUSTOMERS 4. DISPLAY ACCOUNT NO, CUSTOMER NAME & AMOUNT FROM DEPOSIT. SELECT ACTNO, CNAME, AMOUNT FROM DEPOSIT 5. DISPLAY LOAN NO, AMOUNT FROM BORROW table. SELECT LOANNO, AMOUNT FROM BORROW

6. DISPLAY LOAN DETAILS OF ALL CUSTOMERS WHO BELONGS TO 'ANDHERI' BRANCH.
```
SELECT * FROM BORROW WHERE BNAME ='ANDHERI'
```

7. GIVE ACCOUNT NO AND AMOUNT OF DEPOSITOR, WHOSE ACCOUNT NO IS EQUALS TO 106.
```
SELECT ACTNO, AMOUNT FROM DEPOSIT WHERE ACTNO=106
```

8. GIVE NAME OF BORROWERS HAVING AMOUNT GREATER THAN 5000.
```
SELECT CNAME FROM BORROW WHERE AMOUNT>5000
```

9. GIVE NAME OF CUSTOMERS WHO OPENED ACCOUNT AFTER DATE '1-DEC-96'.
```
SELECT CNAME FROM DEPOSIT WHERE ADATE>'1/DEC/96'
```

10. DISPLAY NAME OF CUSTOMERS WHOSE ACCOUNT NO IS LESS THAN 105.
```
SELECT CNAME FROM DEPOSIT WHERE ACTNO<105
```

11. DISPLAY NAME OF CUSTOMER WHO BELONGS TO EITHER 'NAGPUR' OR 'DELHI'. (**OR & IN**)
```
SELECT CNAME FROM CUSTOMERS WHERE CITY='NAGPUR' OR CITY='DELHI'
SELECT CNAME FROM CUSTOMERS WHERE CITY IN('NAGPUR', 'DELHI')
```

12. DISPLAY NAME OF CUSTOMERS WITH BRANCH WHOSE AMOUNT IS GREATER THAN 4000 AND ACCOUNT NO IS LESS THAN 105.
```
SELECT CNAME, BNAME FROM DEPOSIT WHERE AMOUNT>4000 AND ACTNO<105
```

13. FIND ALL BORROWERS WHOSE AMOUNT IS GREATER THAN EQUALS TO 3000 & LESS THAN EQUALS TO 8000. (**AND & BETWEEN**)
```
SELECT * FROM BORROW WHERE AMOUNT>=3000 AND AMOUNT<=8000
SELECT * FROM BORROW WHERE AMOUNT BETWEEN 3000 AND 8000
```

14. FIND ALL DEPOSITORS WHO DO NOT BELONGS TO 'ANDHERI' BRANCH.
```
SELECT * FROM DEPOSIT WHERE BNAME<>'ANDHERI'
```

15. DISPLAY THE NAME OF BORROWERS WHOSE AMOUNT IS NULL.
```
SELECT * FROM BORROW WHERE AMOUNT IS NULL
```

Part – B

16. DISPLAY ACCOUNT NO, CUSTOMER NAME & AMOUNT OF SUCH CUSTOMERS WHO BELONGS TO 'AJNI', 'KAROLBAGH' OR 'M.G.ROAD' AND ACCOUNT NO IS LESS THAN 104.
```
SELECT ACTNO, CNAME, AMOUNT FROM DEPOSIT WHERE BNAME IN ('AJNI', 'KAROLBAGH',
'M.G.ROAD') AND ACTNO<104
```

17. DISPLAY ALL THE DETAILS OF FIRST FIVE CUSTOMERS.
```
SELECT TOP 5 * FROM CUSTOMERS
```

18. DISPLAY ALL THE DETAILS OF FIRST THREE DEPOSITORS WHO'S AMOUNT IS GREATER THAN 1000.
```
SELECT TOP 3 * FROM DEPOSIT WHERE AMOUNT>1000
```

19. DISPLAY LOAN NO, CUSTOMER NAME OF FIRST FIVE BORROWERS WHO'S BRANCH NAME DOES NOT BELONGS TO 'ANDHERI'.
```
SELECT TOP 5 LOANNO, CNAME FROM BORROW WHERE BNAME!='ANDHERI'
```

20. RETRIEVE ALL UNIQUE CITIES USING DISTINCT. (USE **CUSTOMERS TABLE)**

```
SELECT DISTINCT CITY FROM CUSTOMERS
```

Part – C

21. RETRIEVE ALL UNIQUE BRANCHES USING DISTINCT. **(**USE **BRANCH TABLE)**
```
SELECT DISTINCT BNAME FROM BRANCH
```

22. RETRIEVE ALL THE RECORDS OF CUSTOMER TABLE AS PER THEIR CITY NAME IN ASCENDING ORDER.
```
SELECT * FROM CUSTOMERS ORDER BY CITY;
```

23. RETRIEVE ALL THE RECORDS OF DEPOSIT TABLE AS PER THEIR AMOUNT COLUMN IN DESCENDING ORDER.
```
SELECT * FROM DEPOSIT ORDER BY AMOUNT DESC;
```

24. RETRIEVE ALL THE DETAILS OF CUSTOMERS IN DESCENDING ORDER OF THEIR CITY NAME.
```
SELECT * FROM CUSTOMERS ORDER BY CITY DESC;
```

25. SHOW ALL UNIQUE BORROWERS & LABLE THE COLUMN UNI_Borrowers. (Display only Names).
```
SELECT DISTINCT CNAME AS UNI_Borrowers FROM BORROW;
```

5

Implementation of the UPDATE command in SQL
From the tables given in Lab-3, perform the following queries using the UPDATE command:

Part – A

1. UPDATE DEPOSIT AMOUNT OF ALL CUSTOMERS FROM 3000 TO 5000.
```
UPDATE DEPOSIT SET AMOUNT=5000 WHERE AMOUNT=3000
```

2. CHANGE BRANCH NAME OF ANIL FROM VRCE TO C.G.ROAD. (USE **BORROW TABLE**)
```
UPDATE BORROW SET BNAME='C.G.ROAD'WHERE CNAME='ANIL'
```

3. UPDATE ACCOUNT NO OF SANDIP TO 111 & AMOUNT TO 5000.
```
UPDATE DEPOSIT SET ACTNO=111, AMOUNT=5000 WHERE CNAME='SANDIP'
```

4. GIVE 10% INCREMENT IN LOAN AMOUNT.
```
UPDATE BORROW SET AMOUNT=AMOUNT+(AMOUNT*10/100)
```

5. UPDATE DEPOSIT AMOUNT OF ALL DEPOSITORS TO 5000 WHOSE ACCOUNT NO BETWEEN 103 & 107.
```
UPDATE DEPOSIT SET AMOUNT=5000 WHERE ACTNO BETWEEN 103 AND 107
```

Part – B

6. UPDATE AMOUNT OF LOAN NO 321 TO NULL.
```
UPDATE BORROW SET AMOUNT=NULLWHERE LOANNO=321
```

7. CHANGE LOAN NUMBER FROM 201 TO 401 & ALSO CHANGE BRANCH NAME FROM VRCE TO AJNI.
```
UPDATE BORROW SET LOANNO=401, BNAME='AJNI' WHERE LOANNO=201 AND
BNAME='VRCE'
```

8. MODIFY CUSTOMER NAME ANIL TO ANIL JAIN.
```
UPDATE CUSTOMERS SET CNAME='ANIL JAIN' WHERE CNAME='ANIL'
```
9. GIVE 1000 RS INCREMENT IN LOAN AMOUNT.
```
UPDATE BORROW
SET AMOUNT = 1000+AMOUNT;
```
10. UPDATE CUSTOMER NAME MINU TO SINU AND AMOUNT 10000.
```
UPDATE DEPOSIT
SET CNAME = 'SINU'
WHERE AMOUNT = 10000  AND CNAME = 'MINU';
```

Part – C

11. CHANGE NAME TO RAMESH, BRANCH NAME VRCE & AMOUNT 5500, WHOSE ACCOUNT NUMBER IS 102.
```
UPDATE DEPOSIT
SET CNAME = 'RAMESH' , BNAME = 'VRCE' , AMOUNT = 5500
WHERE ACTNO = 102
```
12. MAKE BRANCH NAME & AMOUNT NULL, WHOSE LOAN NUMBER IS 481 & NAME IS KRANTI.
```
UPDATE BORROW
SET BNAME = NULL, AMOUNT = NULL
WHERE LOANNO = 481 AND CNAME = 'KRANTI';
```
13. GIVE 5% INCREMENT IN LOAN AMOUNT WHOSE LOAN NO LESS THAN 321.
```
UPDATE BORROW
SET AMOUNT = 1.05 * AMOUNT
WHERE LOANNO < 321;
```
14. UPDATE THE CUSTOMER CITY BOMBAY TO MUMBAI.
```
UPDATE BRANCH
SET CITY = 'MUMBAI'
WHERE CITY = 'BOMBAY';
```
15. UPDATE THE BRANCH NAME VRCE WHERE LOAN NUMBER 375.
```
UPDATE BORROW
SET BNAME = 'VRCE'
WHERE LOANNO = 375;
```

6

Implementation of DELETE, TRUNCATE and DROP commands in SQL
From the tables given in Lab-3, perform the following queries using the DELETE, TRUNCATE and DROP commands:

Part - A

1. DELETE RECORDS OF CUSTOMER TABLE THAT BELONGS TO BOMBAY CITY.
```
DELETE  FROM CUSTOMERS
WHERE CITY = 'BOMBAY';
```
2. DELETE ALL ACCOUNT NUMBERS WHOSE AMOUNT LESS THAN EQUALS TO 1000.
```
DELETE  FROM DEPOSIT
WHERE AMOUNT <= 1000;
```

3. DELETE BORROWERS WHOSE BRANCH NAME IS 'AJNI'.
```sql
DELETE  FROM BORROW
WHERE BNAME = 'AJNI';
```

4. DELETE ALL THE BORROWERS WHOSE LOAN NUMBER BETWEEN 201 TO 210.
```sql
DELETE  FROM BORROW
WHERE LOANNO BETWEEN 201 AND 210;
```

5. DELETE CUSTOMERS WHO OPENED ACCOUNT AFTER DATE '01-DEC-96'. (USE DEPOSIT TABLE)
```sql
DELETE  FROM DEPOSIT
WHERE ADATE >'1996-12-01'
```

Part - B

6. DELETE ALL THE RECORDS OF CUSTOMERS TABLE. (USE TRUNCATE)
```sql
TRUNCATE TABLE CUSTOMERS;
```

7. REMOVE ALL CUSTOMERS WHOSE NAME IS ANIL & ACCOUNT NUMBER IS 101.
```sql
DELETE FROM DEPOSIT
WHERE CNAME = 'ANIL' AND ACTNO = 101;
```

8. DELETE ALL THE DEPOSITORS WHO DO NOT BELONGS TO 'ANDHERI' BRANCH.
```sql
DELETE FROM DEPOSIT
WHERE BNAME != 'ANDHERI';
```

9. DELETE LOAN DETAILS OF CUSTOMERS WHOSE AMOUNT IS LESS THAN 3000.
```sql
DELETE  FROM BORROW
WHERE AMOUNT<3000;
```

10. DELETE CUSTOMERS WHO OPENED ACCOUNT BEFORE DATE '1-JAN-96'. (USE DEPOSIT TABLE)
```sql
DELETE  FROM DEPOSIT
WHERE ADATE<'1996-01-01';
```

Part - C

11. DELETE ALL THE BORROWERS WHOSE LOAN AMOUNT IS LESS THAN 2000 AND DOES NOT BELONG TO VRCE BRANCH.
```sql
DELETE FROM BORROW
WHERE AMOUNT<2000 AND BNAME!= 'VRCE';
```

12. DELETE ALL THE RECORDS OF DEPOSIT TABLE. (USE TRUNCATE)
```sql
TRUNCATE TABLE DEPOSIT;
```

13. DELETE ALL THE RECORDS OF BRANCH TABLE. (USE TRUNCATE)
```sql
TRUNCATE TABLE BRANCH;
```

14. REMOVE DEPOSIT TABLE. (USE DROP)
```sql
DROP TABLE DEPOSIT;
```

15. REMOVE BRANCH TABLE. (USE DROP)
```sql
DROP TABLE BRANCH;
```

7 Implementation of the LIKE Operator in SQL

Perform the following queries on the above-given table:

Part – A

1. DISPLAY THE NAME OF STUDENTS WHOSE NAME STARTS WITH ‘K’.
   `SELECT FIRSTNAME FROM STUDENT WHERE FIRSTNAME LIKE 'K%'`

2. DISPLAY THE NAME OF STUDENTS WHOSE NAME CONSISTS OF FIVE CHARACTERS.
   `SELECT FIRSTNAME FROM STUDENT WHERE FIRSTNAME LIKE '_____'`

3. RETRIEVE THE FIRST NAME & LAST NAME OF STUDENTS WHOSE CITY NAME ENDS WITH A & CONTAINS SIX CHARACTERS.
   `SELECT FIRSTNAME, LASTNAME, CITY FROM STUDENT WHERE CITY LIKE'_____A'`

4. DISPLAY ALL THE STUDENTS WHOSE LAST NAME ENDS WITH ‘TEL’.
   `SELECT * FROM STUDENT WHERE LASTNAME LIKE '%TEL'`

5. DISPLAY ALL THE STUDENTS WHOSE FIRST NAME STARTS WITH ‘HA’ & ENDS WITH ‘T’.
   `SELECT * FROM STUDENT WHERE FIRSTNAME LIKE 'HA%T'`

6. DISPLAY ALL THE STUDENTS WHOSE FIRST NAME STARTS WITH ‘K’ AND THIRD CHARACTER IS ‘Y’.
   `SELECT * FROM STUDENT WHERE FIRSTNAME LIKE 'K_Y%'`

7. DISPLAY THE NAME OF STUDENTS HAVING NO WEBSITE AND NAME CONSISTS OF FIVE CHARACTERS.
   `SELECT FIRSTNAME FROM STUDENT WHERE WEBSITE IS NULL AND FIRSTNAME LIKE'_____'`

8. DISPLAY ALL THE STUDENTS WHOSE LAST NAME CONSISTS OF ‘JER’.
   `SELECT * FROM STUDENT WHERE LASTNAME LIKE '%JER%'`

9. DISPLAY ALL THE STUDENTS WHOSE CITY NAME STARTS WITH EITHER ‘R’ OR ‘B’.
   `SELECT * FROM STUDENT WHERE CITY LIKE 'R%' OR CITY LIKE 'B%'`

10. DISPLAY ALL THE STUDENTS NAME HAVING WEBSITES.
    `SELECT FIRSTNAME, LASTNAME FROM STUDENT WHERE WEBSITE LIKE '%'`

Part – B

11. DISPLAY ALL THE STUDENTS WHOSE NAME STARTS FROM ALPHABET A TO H.
    `SELECT * FROM STUDENT WHERE FIRSTNAME LIKE '[A-H] %'`

12. DISPLAY ALL THE STUDENTS WHOSE NAME’S SECOND CHARACTER IS VOWEL.
    `SELECT * FROM STUDENT WHERE FIRSTNAME LIKE '_[AEIOU]%'`

13. DISPLAY STUDENT’S NAME WHOSE CITY NAME CONSIST OF ‘ROD’.
    `SELECT FIRSTNAME FROM STUDENT WHERE CITY LIKE '%ROD%'`

14. RETRIEVE THE FIRST & LAST NAME OF STUDENTS WHOSE WEBSITE NAME STARTS WITH ‘BI’.
    `SELECT FIRSTNAME, LASTNAME FROM STUDENT WHERE WEBSITE LIKE 'BI%'`

15. DISPLAY STUDENT'S CITY WHOSE LAST NAME CONSISTS OF SIX CHARACTERS.
    SELECT CITY FROM STUDENT WHERE LASTNAME LIKE '______'

Part – C

16. DISPLAY ALL THE STUDENTS WHOSE CITY NAME CONSIST OF FIVE CHARACTERS & NOT START WITH 'BA'.
    SELECT * FROM STUDENT WHERE CITY LIKE '_____' AND CITY NOT LIKE 'BA%'

17. SHOW ALL THE STUDENT'S WHOSE DIVISION STARTS WITH 'II'.
    SELECT * FROM STUDENT WHERE DIVISION LIKE 'II%'

18. FIND OUT STUDENT'S FIRST NAME WHOSE DIVISION CONTAINS 'BC' ANYWHERE IN DIVISION NAME.
    SELECT FIRSTNAME, DIVISION FROM STUDENT WHERE DIVISION LIKE '%BC%'

19. SHOW STUDENT ID AND CITY NAME IN WHICH DIVISION CONSIST OF SIX CHARACTERS AND HAVING WEBSITE NAME.
    SELECT STUID, CITY FROM STUDENT WHERE DIVISION LIKE '______' AND WEBSITE IS NOT NULL

20. DISPLAY ALL THE STUDENTS WHOSE NAME'S THIRD CHARACTER IS CONSONANT.
    SELECT * FROM STUDENT WHERE FIRSTNAME NOT LIKE '__[AEIOU]%'

8

Implementation of the ALTER and RENAME commands in SQL
Create the given table using Query Mode and perform the following queries.

Part – A

1. ADD TWO MORE COLUMNS CITY VARCHAR (20) NULL AND BACKLOG INT NOT NULL.
   ALTER TABLE STUDENT ADD CITY VARCHAR(20), BACKLOG INT NOT NULL

2. CHANGE THE SIZE OF NAME COLUMN OF STUDENT FROM VARCHAR (25) TO VARCHAR (35).
   ALTER TABLE STUDENT ALTER COLUMN NAME VARCHAR(35)

3. CHANGE THE DATA TYPE DECIMAL TO INT IN CPI COLUMN.
   ALTER TABLE STUDENT ALTER COLUMN CPI INT

4. RENAME COLUMN ENROLLMENT NO TO ENO.
   SP_RENAME 'STUDENT.ENROLLMENTNO', 'ENO', 'COLUMN'

5. DELETE COLUMN CITY FROM THE STUDENT TABLE.
   ALTER TABLE STUDENT DROP COLUMN CITY

6. CHANGE NAME OF TABLE STUDENT TO STUDENT_MASTER.
   SP_RENAME 'STUDENT', 'STUDENT_MASTER'

Part – B

7. REMOVE COLUMN BACKLOG FROM THE TABLE.
   ALTER TABLE STUDENT DROP COLUMN BACKLOG

8. CHANGE CONSTRAINT OF NAME COLUMN FROM NULL TO NOT NULL.
   ALTER TABLE STUDENT ALTER COLUMN NAME VARCHAR(25) NOT NULL

Part – C

9. RENAME COLUMN BIRTHDATE TO BDATE.

```
SP_RENAME 'STUDENT.BIRTHDATE', 'BDATE', 'COLUMN'
```

10. CHANGE THE DATATYPE OF ENO COLUMN FROM VARCHAR (20) TO VARCHAR (12)

```
ALTER TABLE STUDENT ALTER COLUMN ENO VARCHAR(12)
```